Research Ethics Application Form (Part A & B)

School of Computing · Agile Team Project (ATP — 5COSC021W)

Ethics Ref: CS-43-2026

CW1 Portfolio Chapter 5

PART A: GENERAL PROJECT & INVESTIGATOR DETAILS

Project Title	**ALVI: Autonomous Home-Care Companion Robot & Caregiver Web Application**		
Module & Coursework	5COSC021W Agile Team Project (CW1/CW2)	**Team ID & Board**	Team 43 (Jira: SCRUM)
Academic Supervisor	Faculty Supervisor (School of Computing, IIT)	**Target Submission**	November 2026 (CW1)

STUDENT RESEARCH INVESTIGATORS

Full Student Name	UoW ID	IIT Student ID	Project Role & Responsibility
Amodh Gunawardana	W22133081	20250165	Product Owner / Tech Architect / Elicitation Lead
Nethum Liyanage	W2212876	20250172	Scrum Master / Ethics & Testing Lead / Backend
Thisara Chandima	W2212870	20250166	Hardware Lead / Embedded Sensors & Physical Safety
Thevindu Herath	W2212837	20250157	Lead UI/UX Designer / Accessible Patient Interfaces
Sithil Hettiarachchi	W2212867	20250162	Portfolio Documentation Lead / SLEP & Traceability

PART A: ETHICAL RISK SCREENING CHECKLIST

Screening items answered 'YES' trigger full Part B mitigation statements below:

Ethical Consideration Checklist Question	YES	NO
1. Does the study involve direct interaction with human participants (e.g. interviews, surveys, evaluations)?	☑	☐
2. Does the project involve vulnerable participants (e.g. elderly persons living alone, patients with chronic ailments)?	☑	☐
3. Will identifiable personal or health-related data be collected (e.g. heart rate, medication routines, vitals)?	☑	☐
4. Does the project involve audio/video recordings during research sessions?	☑	☐
5. Does the project involve testing physical robotic hardware or electrical actuators around humans?	☑	☐
6. Are research findings and user personas published in student portfolios and public demonstrations?	☑	☐
7. Will financial incentives, commercial payments, or copyright assignments be offered to participants?	☐	☑
8. Is there any potential for psychological harm, discomfort, deception, or invasive physical procedures?	☐	☑

Mandatory Ethics Trigger (Part B Required): Items 1 through 6 are flagged affirmative due to field elicitation involving healthcare specialists and elderly subjects, audio transcription, health vital streams, and physical assistive robotics. Part B provides full ethical protocols under **Sri Lanka PDPA No. 9 of 2022**, **UK GDPR**, and the **BCS Code of Conduct**.

Part B: Ethical Considerations & Risk Mitigation Protocol

Project ALVI — Autonomous Companion Robot (Jira: SCRUM-61 / ST-09.2)

1. VULNERABLE PARTICIPANT SAFEGUARDS (ELDERLY & PATIENTS)

To protect elderly participants and care recipients from exploitation or privacy breaches, all interviews strictly enforce **Option B (Mandatory Anonymity)**. Real patient names, residential addresses, and exact medical diagnoses are permanently withheld. Transcripts are coded under academic pseudonyms (e.g. `[Subject: Patient-01]`). Formal legal guardians or family caregivers are permitted to attend sessions upon request.

2. INFORMED CONSENT & RETROSPECTIVE WITHDRAWAL PROTOCOL

All participants receive the official *Participant Informed Consent & Recognition Form (Doc Ref: ALVI-ETH-01-V2)* prior to engagement. Consent is explicit, written, and recorded with 6 affirmative checkpoints. Participants maintain the unqualified right to pause or terminate the session. A strict **retrospective withdrawal cut-off date** is enforced: participants may withdraw data within **7 calendar days post-interview or before 20 November 2026**, after which findings are irreversibly synthesized into portfolio deliverables.

3. DATA SECURITY, PRIVACY & DESTRUCTION POLICY (PDPA 2022 & UK GDPR)

Under Section 10 of the Sri Lanka Personal Data Protection Act No. 9 of 2022 and Article 5(1)(e) of UK GDPR: (a) All audio/video recordings and raw notes are stored in an AES-256 encrypted storage volume restricted by two-factor authentication to Team 43 leads. (b) Live vitals (heart rate BPM) and vision AI streams are processed locally in real-time on the Tier 3 edge laptop without remote storage. (c) **Permanent Secure Destruction:** All raw recordings will be permanently purged and shredded within 30 days of module exam board ratification (no later than **31 May 2027**).

4. HARDWARE SAFETY & PROTOTYPING RISK ASSESSMENT

The robotic mobile base operates strictly at safe low-voltage DC (12V SLA battery and 5V USB step-downs). Pill dispenser cylinders are fabricated from non-toxic, food-grade PLA filament. Hardware includes a physical emergency power kill-switch, ultrasonic and LiDAR collision stop buffers (minimum 25cm safety radius), and optical IR break-beam verification to avoid dispenser pinch risks.

5. FORMAL ETHICS DECLARATIONS & SIGNATURES

Student Ethics & Scrum Master Lead

I declare that the information provided in Parts A & B is true and accurate, and that research activities will be conducted in full compliance with UoW and BCS ethical codes.

Signature: Nethum Liyanage (Scrum Master / Ethics Lead)

Date: 09 / 11 / 2026 **UoW ID:** W2212876

Academic Supervisor / Faculty Ethics Sign-off

I confirm that I have reviewed the ethics application for Team 43 (ALVI). The proposed elicitation methodologies and safeguards meet university ethical standards.

Supervisor Signature (School of Computing, IIT)

Date: ____ / ____ / 2026 **Decision:** [ ] APPROVED